

EXPERIMENT NO 5

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Div:- SY3

Batch:- C3

INPUT CODE

```
#include <stdio.h>
#include <stdlib.h>
#define size 5

struct stack
{
    int s[size];
    int top;
} st;

int stfull()
{
    if (st.top >= size - 1)
        return 1;
    else
        return 0;
}

int stempty()
{
    if (st.top == -1)
        return 1;
    else
        return 0;
}

void push(int item)
{
```

```

    st.top++;
    st.s[st.top] = item;
}

int pop()
{
    int item;

    item = st.s[st.top];
    st.top--;

    return item;
}

void displaystack()
{
    int i;

    if (st.empty())
    {
        printf("\nThe Stack is empty");
    }
    else
    {
        for (i = st.top; i >= 0; i--)
        {
            printf("\n%d", st.s[i]);
        }
    }
}

int main()
{
    int item, choice;
    char ans;

    st.top = -1;

```

```
printf("\nImplementation of Stack\n");
```

```
do
```

```
{
```

```
    printf("\n\nMain Menu");
```

```
    printf("\n1. Push");
```

```
    printf("\n2. Pop");
```

```
    printf("\n3. Display");
```

```
    printf("\n4. Exit");
```

```
    printf("\n\nEnter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice)
```

```
    {
```

```
        case 1:
```

```
            printf("\nEnter the item to be pushed: ");
```

```
            scanf("%d", &item);
```

```
            if (stfull())
```

```
            {
```

```
                printf("\nStack is full");
```

```
            }
```

```
            else
```

```
            {
```

```
                push(item);
```

```
            }
```

```
            break;
```

```
        case 2:
```

```
            if (stempty())
```

```
            {
```

```
                printf("\nThe Stack is empty");
```

```
            }
```

```
            else
```

```
            {
```

```
        item = pop();
        printf("\nThe popped element is %d", item);
    }
    break;

case 3:
    displaystack();
    break;

case 4:
    printf("\nExiting program...");
    return 0;

default:
    printf("\nInvalid choice!");
}

printf("\n\nDo you want to continue? (Y/N): ");
scanf(" %c", &ans);

} while (ans == 'Y' || ans == 'y');

return 0;
}
```

OUTPUT

Implementation of Stack

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the item to be pushed: 10

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the item to be pushed: 20

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the item to be pushed: 30

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

30

20

10

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 2

The popped element is 30

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

20

10

Do you want to continue? (Y/N): 4

Implementation of Stack

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 2

The Stack is empty

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

The Stack is empty

Do you want to continue? (Y/N): y

Main Menu

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 4

Exiting program...

...Program finished with exit code 0
Press ENTER to exit console.